



Unit-1

Database system Architecture





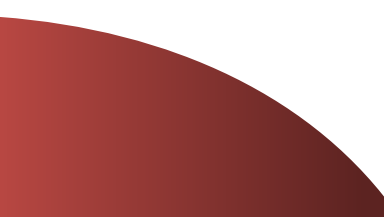
Outline

- Introduction of DBMS
- Applications of DBMS
- Advantages of DBMS
- Three levels ANSI SPARC database system
- Data Abstraction in DBMS
- Mappings and data independence
- Database users and DBA
- Database system architecture



Introduction to DBMS

Section - 1



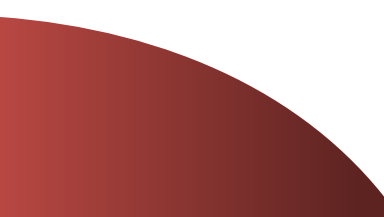
What is Database Management System (DBMS)?

- ▶ Data - **Fact** that can be recorded or stored
 - ↳ e.g. Person Name, Age, Gender and Weight etc.
- ▶ Database - Collection of **logically related data**
 - ↳ e.g. Books Database in Library, Student Database in University etc.
- ▶ Management - Manipulation, Searching and Security of data
 - ↳ e.g. Viewing result in GTU website, Searching exam papers in GTU website etc.
- ▶ System - **Programs** or **tools** used to manage database
 - ↳ e.g. SQL Server Studio Express, Oracle etc.
- ▶ DBMS - A Database Management System is a software for creating and managing databases.
- ▶ Database Management System (DBMS) is a **software designed to define, manipulate, retrieve and manage data in a database.**
 - ↳ e.g. MS SQL Server, Oracle, My SQL, SQLite, MongoDB etc.



Applications of DBMS

Section - 2



Applications of DBMS

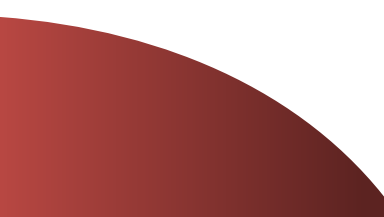
- ▶ DBMS is a computerized record-keeping system.
- ▶ DBMS is required where ever data need to be stored.
 - ➔ E-Commerce (**Flikart, Amazon, Shopclues, eBay** etc...)
 - ➔ Online Television Streaming (**Hotstar, Amazon Prime** etc...)
 - ➔ Social Media (**WhatsApp, Facebook, Twitter, LinkedIn** etc...)
 - ➔ Banking & Insurance
 - ➔ Airline & Railway
 - ➔ Universities and Colleges/Schools
 - ➔ Library Management System
 - ➔ Human Resource Department
 - ➔ Hospitals and Medical Stores
 - ➔ Government Organizations

Exercise Write down any five applications of DBMS other than above.



Advantages of DBMS

Section - 3



Reduce data redundancy (duplication)

Computer

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	1234	DBMS

Database management system can remove such data redundancy by storing data centrally.

Civil

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	1234	DBMS

Same data is stored at four different places.



Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	1234	DBMS

Electrical

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	1234	DBMS

Mechanical

Remove data inconsistency

Computer

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	6789	DBMS

Database management system can keep data in consistent state.

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	6789	DBMS

Electrical

Civil

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	1234	DBMS

Same data having different state (values)

Mobile no is changed

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	1234	DBMS

Mechanical

Data isolation

- ▶ Data are **scattered** in various files.
- ▶ Files may be in **different formats**.
- ▶ **Difficult to retrieve** the appropriate data.

DBMS allow us to access (retrieve) appropriate data easily.

Data isolation is a property that determines when and how changes made by one operation become visible to other concurrent users and systems.

This issue occurs in a concurrency situation.

File - 1

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	1234	DBMS

File - 2

Emp_Name	Post	Salary	Load
Prof. john wick	Lecturer	50,000	15

File - 3

Emp_Name	Teaching	Knowledge	Rating
Prof. john wick	Good	Excellent	9

Guaranteed atomicity

- ▶ Atomicity: Either transaction **execute 0% or 100%**.



Person A
Account A
Bal : 2000

Sum of both account
before transfer is 3000



Person B
Account B
Bal : 1000

Transfer 500

Step 1 : Debit 500 from Account A
Step 2 : Credit 500 into Account B

Transaction
is failed

Sum of both account
after transfer is 3000

Sum of both
account is 2500
so inconsistent

Allow to implement integrity constraints

Emp_Name	Address	Mobile_No	Subject
Prof. john wick	Newyork	9876543210	DBMS

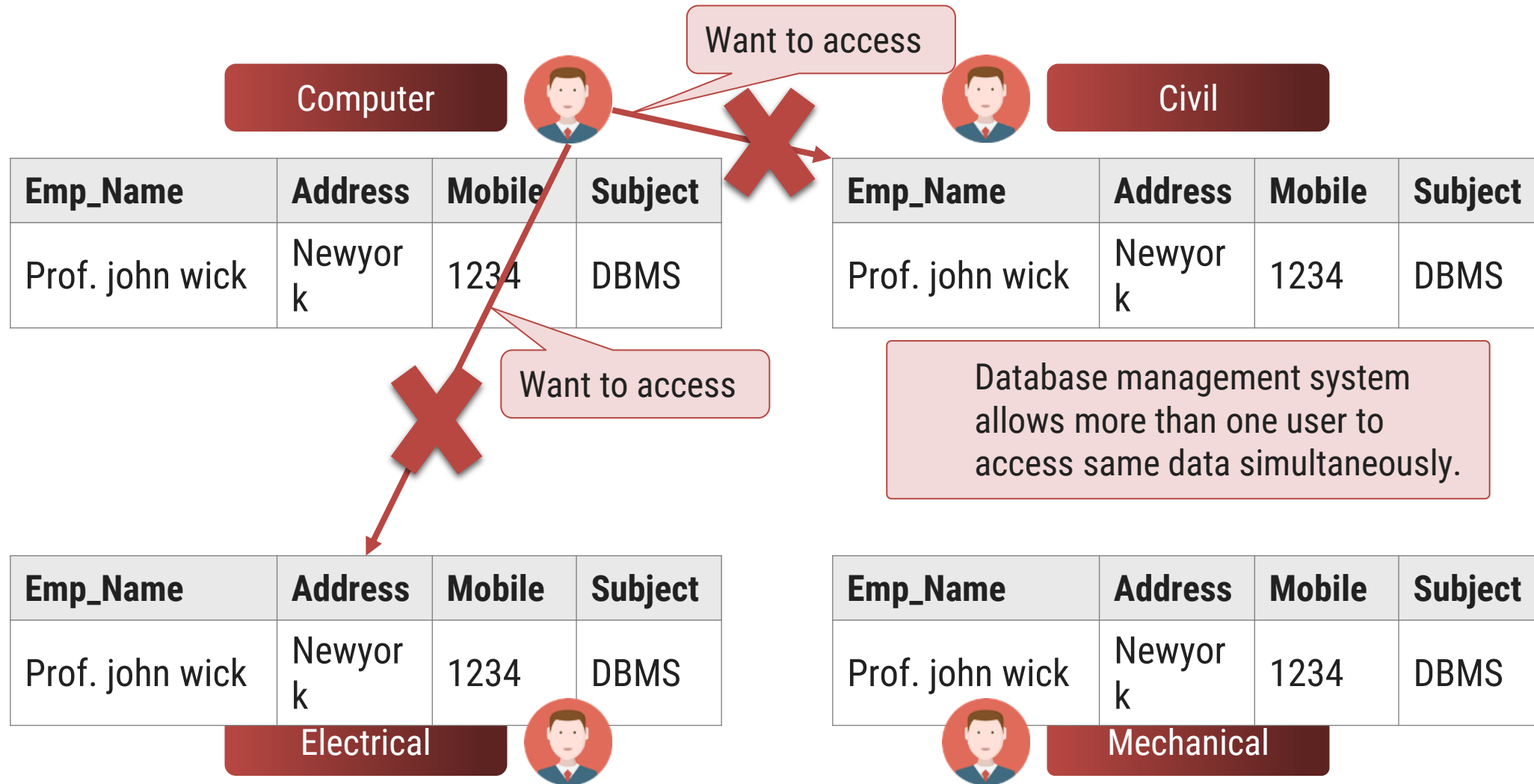
Should contain exact 10 digits

Student_Name	Branch	Backlog	SPI
Nirav Patel	Newyork	0	8.5

Should be between 0 to 10

DBMS allows us to implement such business rules in our database.

Sharing of data among multiple users



Restricting unauthorized access to data

File - 1

Emp_Name	Address	Mobile	Subject
Prof. john wick	Newyork	1234	DBM ^c

File - 2

Emp_Name	Post	Salary	Load
Prof. john wick	Lecturer	50,000	15



File - 3

Emp_Name	Teaching	Knowledge	Rating
Prof. john wick	Good	Excellent	9

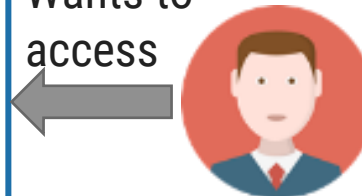


Wants to access



Faculty of other college

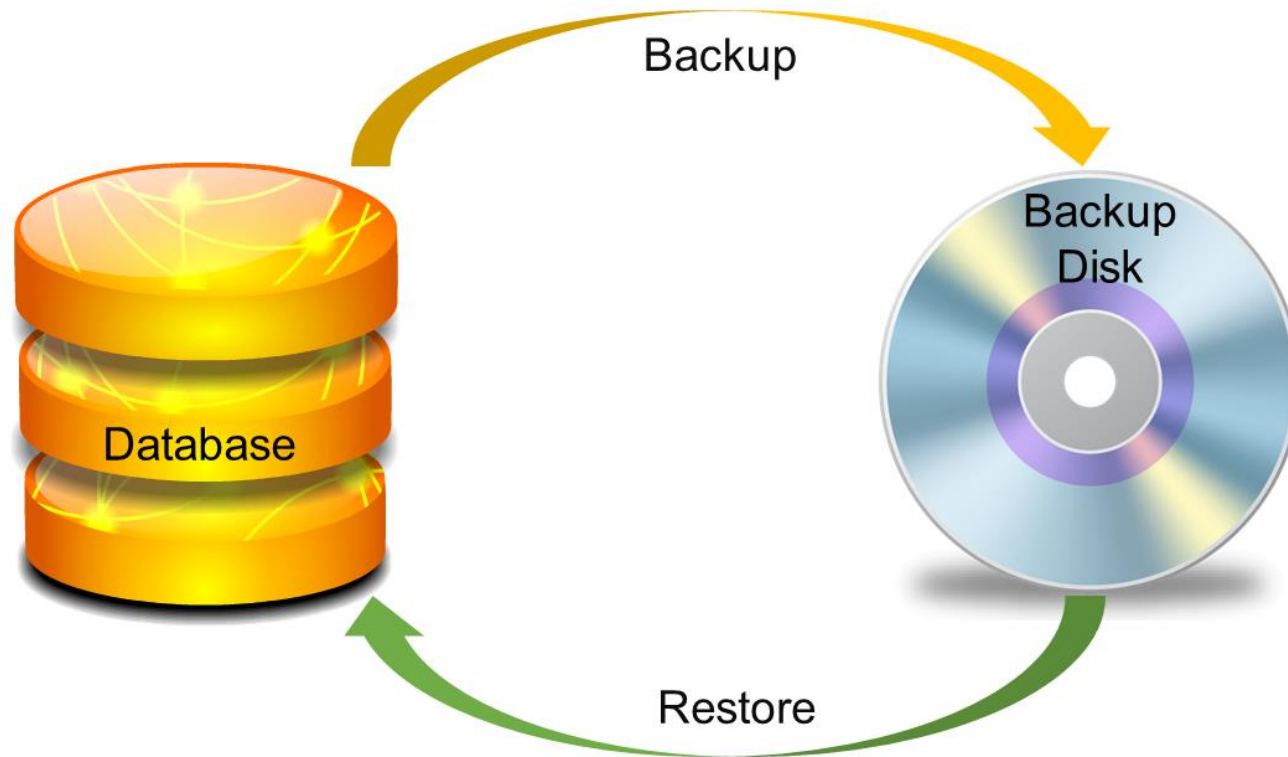
Wants to access



ITM Faculty

DBMS prevents unauthorized user to access data.

Providing backup and recovery services



Provides facilities to backup and restore the database in case of failure.

Advantages of DBMS (Summary)

- ▶ Reduce data redundancy (duplication)
 - ↳ **Avoids unnecessary duplication** of data by storing data centrally.
- ▶ Remove data inconsistency
 - ↳ By **eliminating redundancy**, data **inconsistency can be removed**.
- ▶ Data isolation
 - ↳ A user can **easily retrieve proper data** as per his/her requirement.
- ▶ Guaranteed atomicity
 - ↳ Either transaction **executes 0% or 100%**.

Advantages of DBMS (Summary)

- ▶ Allow implementing integrity constraints
 - ↳ **Business rules can be implemented** such as do not allow to store amount less than Rs. 0 in balance.
- ▶ Sharing of data among multiple users
 - ↳ **More than one users can access** same data at the same time.
- ▶ Restricting unauthorized access to data
 - ↳ A user can **only access data which is authorized** to him/her.
- ▶ Providing backup and recovery services
 - ↳ Can **take a regular auto or manual backup** and **use it to restore** the database if it corrupts.

File Versus a DBMS

Difference between File System and DBMS

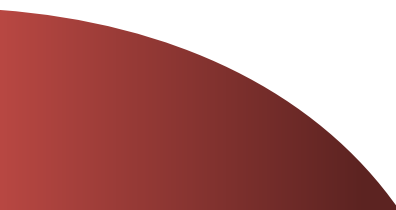
Basics	File System	DBMS
Structure	The file system is a way of arranging the files in a storage medium within a computer.	DBMS is software for managing the database.
Data Redundancy	Redundant data can be present in a file system.	In DBMS there is no redundant data.
Backup and Recovery	It doesn't provide Inbuilt mechanism for backup and recovery of data if it is lost.	It provides in house tools for backup and recovery of data even if it is lost.
Query processing	There is no efficient query processing in the file system.	Efficient query processing is there in DBMS.
Consistency	There is less data consistency in the file system.	There is more data consistency because of the process of normalization.

Security Constraints	File systems provide less security in comparison to DBMS.	DBMS has more security mechanisms as compared to file systems.
Cost	It is less expensive than DBMS.	It has a comparatively higher cost than a file system.
Data Independence	There is no data independence.	In DBMS data independence exists, mainly of two types: 1) Logical Data Independence. 2) Physical Data Independence.
User Access	Only one user can access data at a time.	Multiple users can access data at a time.
Meaning	The users are not required to write procedures.	The user has to write procedures for managing databases
Sharing	Data is distributed in many files. So, it is not easy to share data.	Due to centralized nature data sharing is easy



Basic Terms

Section - 4



Basic terms

► Data

- Data is **raw, unorganized facts** that need to be processed.
- Example: Marks of students
- Student_1 = 50/100, Student_2 = 25/100.

► Information

- When data is **processed, organized, structured** or presented in a given context so as to make it useful, it is called information.
- Example: Result of students (Pass or Fail)
- Student_1 = Pass, Student_2 = Fail.

Basic terms (cont...)

► Metadata

- ➔ Metadata is **data about data**.
- ➔ Data such as table name, column name, data type, authorized user and user access privileges for any table is called metadata for that table.

Faculty			
Emp_Name	Address	Mobile_No	Subject
Prof. john wick	Newyork	9876543210	DBMS

- ➔ Metadata of above table is:
 - Table name such as **Faculty**
 - Column name such as **Emp_Name, Address, Mobile_No, Subject**
 - Datatype such as **Varchar, Decimal**
 - Access privileges such as **Read, Write (Update)**

Basic terms (cont...)

► Data dictionary

→ A data dictionary is an information repository which **contains metadata**.

- Table Name – Faculty
- Column Name – EmpName, Address, Mob, Subject, Salary
- Datatype – Varchar, Decimal
- Access Privileges – Read, Write (Update)

► Data warehouse

→ A data warehouse is an information repository which **stores data**.

Faculty			
Emp_Name	Address	Mobile_No	Subject
Prof. john wick	Newyork	9876543210	DBMS
Prof. Ajay Patel	Surat	0123456789	DBMS

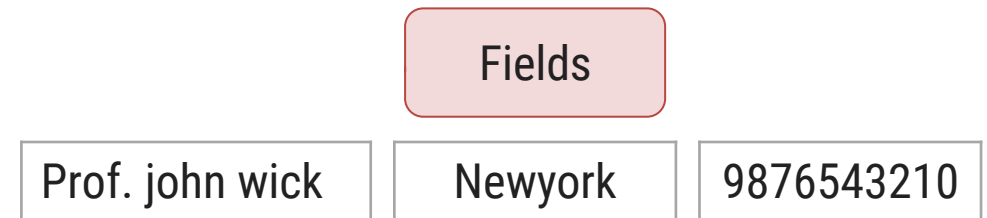
Exercise Why data dictionary and data warehouse are stored in the different places?

Basic terms (cont...)

► Field

- ➔ A field is a **character or group of characters** that have a specific meaning.
- ➔ E.g, the value of Emp_Name, Address, Mobile_No etc are all fields of Faculty table.

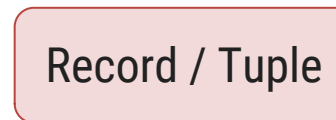
Faculty			
Emp_Name	Address	Mobile_No	Subject
Prof. john wick	Newyork	9876543210	DBMS
Prof. Ajay Patel	Surat	0123456789	DBMS



► Record / Tuple

- ➔ A record is a **collection of logically related fields**.
- ➔ E.g, the collection of fields (Emp_Name, Address, Mobile_No, Subject) forms a record for the Faculty.

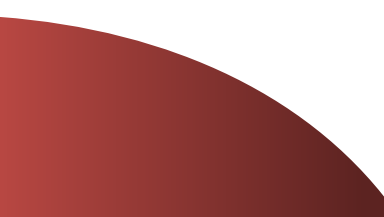
Prof. john wick	Newyork	9876543210	DBMS
Prof. Ajay Patel	Surat	0123456789	DBMS



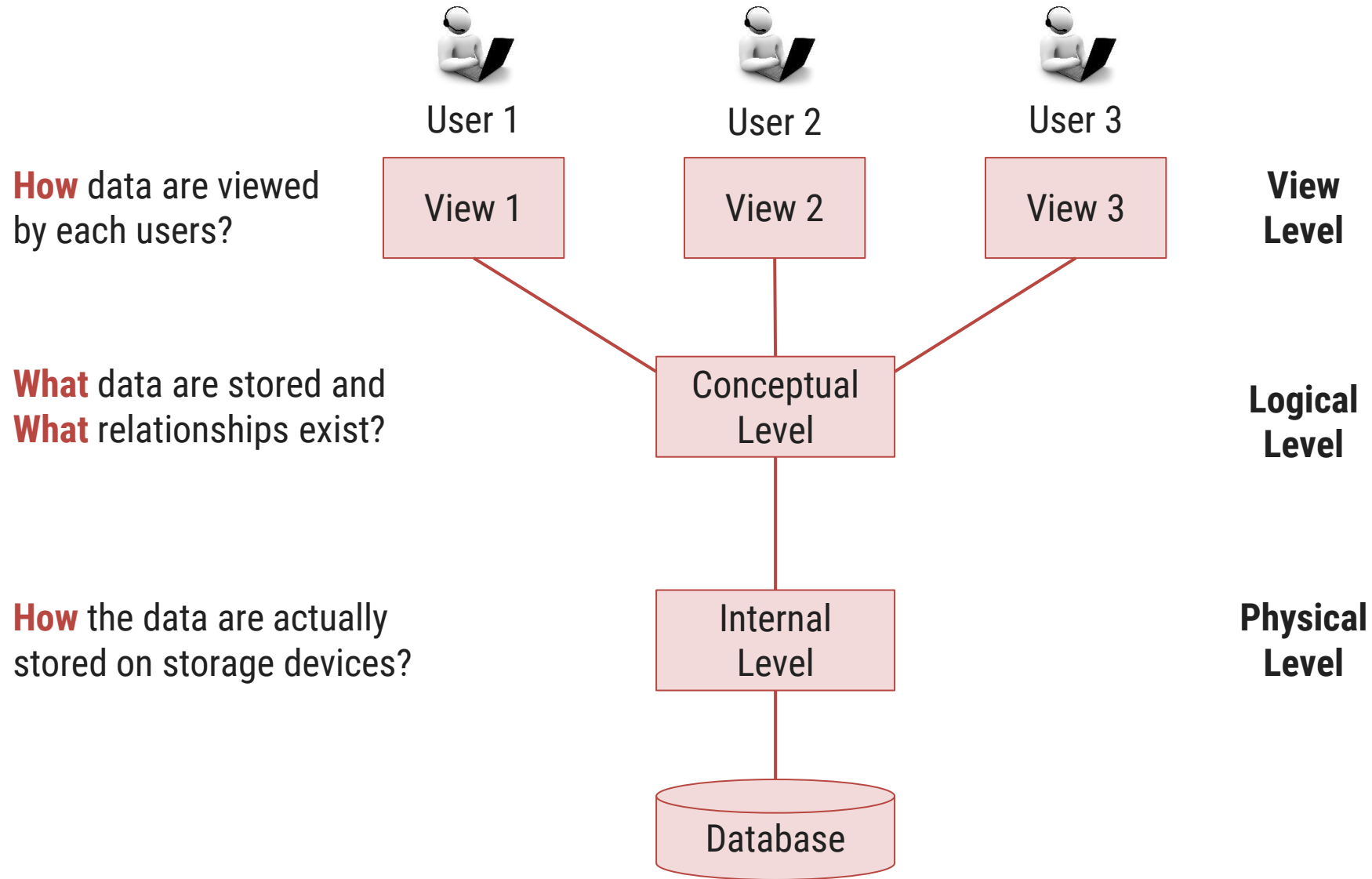


3 Levels Database System

Section - 5



3 Levels Database System



3 Levels ANSI SPARC Database System

▶ Internal level (Physical level)

- It describes **how a data is stored** on the storage device.
- Deals with physical storage of data.
 - Structure of records on disk - files, pages, blocks and indexes and ordering of records
- Internal view is described by the internal schema.

▶ Conceptual level (Logical level)

- **What data are stored and what relationships exist** among those data?
- It hides low level complexities of physical storage.
- For Example, STUDENT database may contain STUDENT and COURSE tables which will be visible to users but users are unaware about their storage.
- Database administrator works at this level to determine what data to keep in the database.

▶ External level (View level)

- It describes only part of the entire database that an end user concern or **how data are viewed** by each user.
- Different user needs different views of the database, so there can be many views in a view level abstraction of the database. Used by end users and application programmers.
- End users need to access only part of the database rather than the entire database.

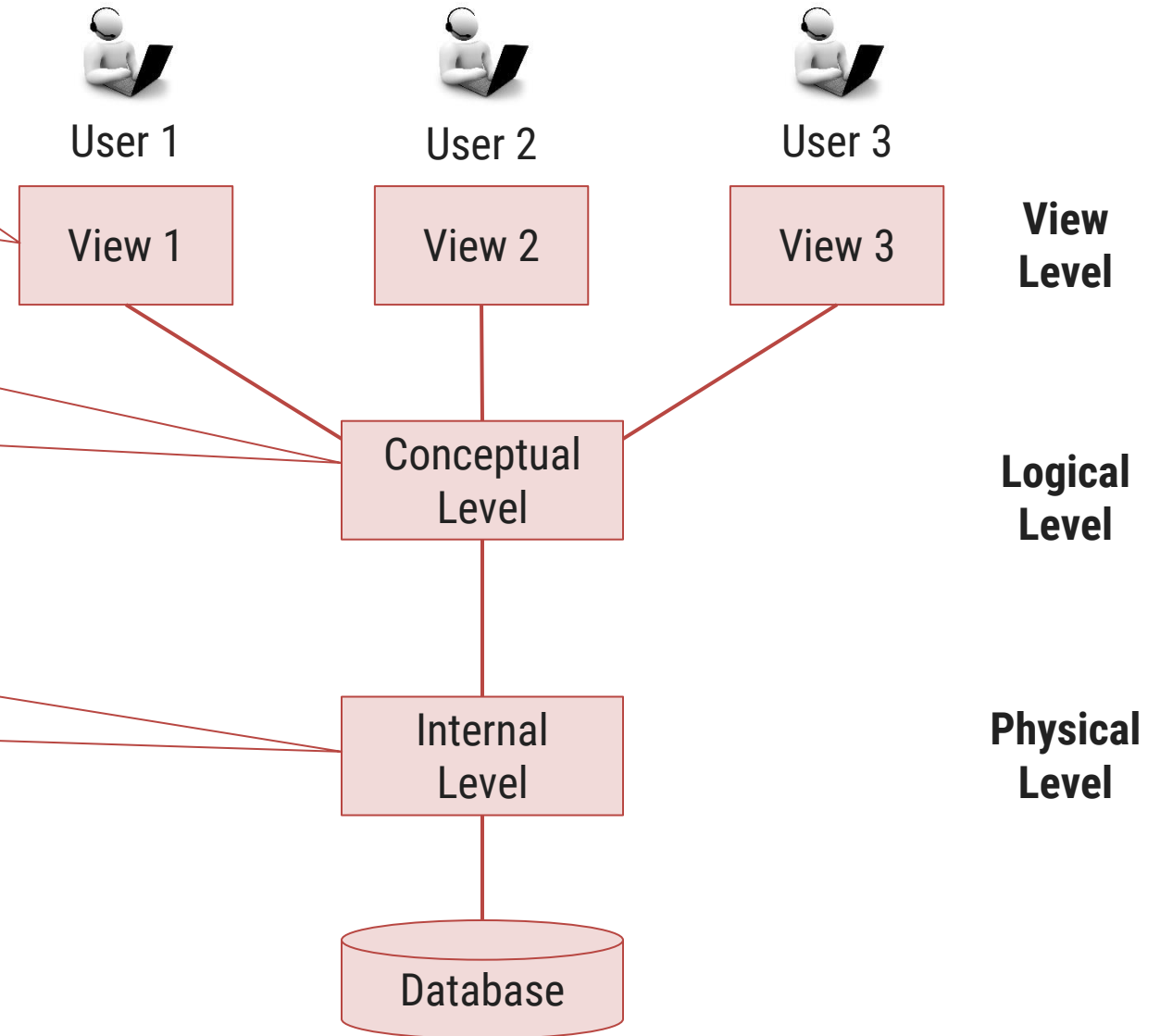
3 Levels ANSI SPARC Database System: Example

We are storing student information in a student table.

User just interact with system with the help of GUI.
Users are not aware of how and what the data is stored.

Records can be described as fields and attributes along
with their data types, their relationship among each other
can be logically implemented.
Programmers generally work at this level.

Records can be described as blocks of storage (bytes,
gigabytes, terabytes etc.) in memory.
These details are often hidden from the programmers.



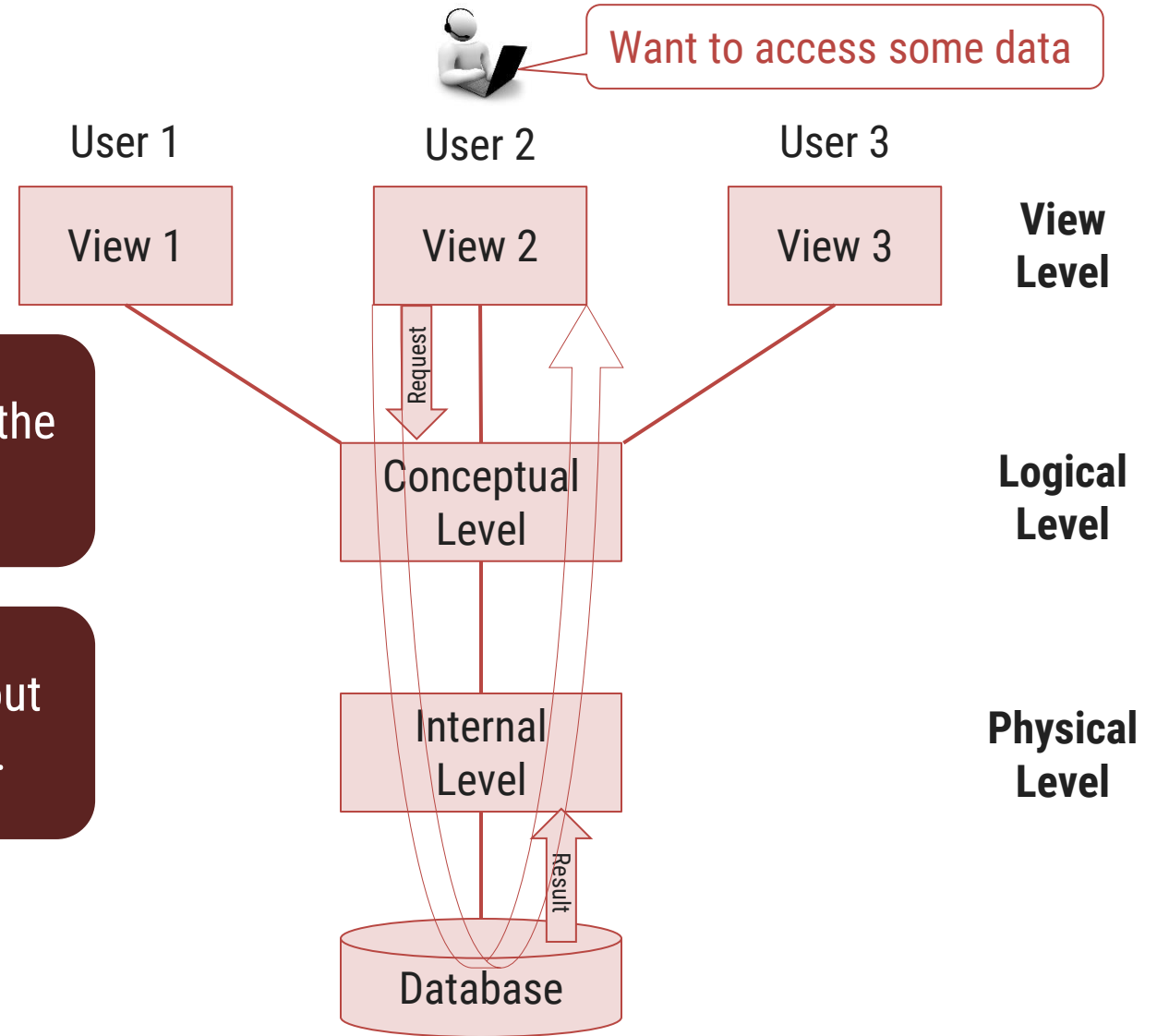
Data Abstraction in DBMS

- ▶ Database systems are made-up of complex data structures.
- ▶ To ease the user interaction with database, the developers hide internal irrelevant details from users.
- ▶ This **process of hiding irrelevant details** from user is called data abstraction.

Mapping and Data Independence

Process of transforming requests and results between the three levels is called mapping.

Ability to modify a schema definition in one level without affecting a schema definition in the next higher level.



Types of Data Independence

▶ Physical Data Independence

- Physical Data Independence is the ability to modify the physical schema without requiring any change in logical (conceptual) schema and application programs.
- Modifications at the internal levels are occasionally necessary to improve performance.
- Possible modifications at internal levels are changes in file structures, compression techniques, hashing algorithms, storage devices, etc.

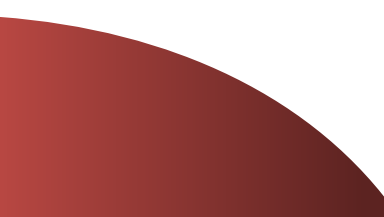
▶ Logical Data Independence

- Logical data independence is the ability to modify the conceptual schema without requiring any change in application programs.
- Modification at the logical levels is necessary whenever the logical structure of the database is changed.
- Application programs are heavily dependent on logical structures of the data they access. So any change in logical structure also requires programs to change.



Types of Database Users

Section - 6



Types of Database Users

▶ Naive Users (End Users)

- ↳ **Unsophisticated users** who have zero knowledge of database system
- ↳ End user interacts to database via sophisticated software or tools
- ↳ e.g. Clerk in bank

▶ Application Programmers

- ↳ **Programmers** who write software using tools such as Java, .Net, PHP etc...
- ↳ e.g. Software developers

▶ Sophisticated Users

- ↳ **Interact with database system** without using an application program
- ↳ Use query tools like SQL
- ↳ e.g. Analyst

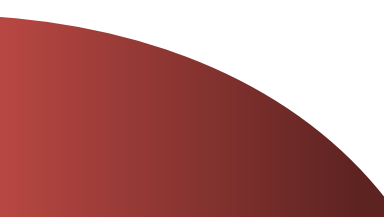
▶ Specialized Users (DBA)

- ↳ User **write specialized** database applications program
- ↳ Use administration tools
- ↳ e.g. Database Administrator



Role of DBA (Database Administrator)

Section - 7



Role of DBA

- ▶ Schema Definition
 - ↳ DBA **defines the logical schema** of the database.
- ▶ Storage Structure and Access Method Definition
 - ↳ DBA **decides how the data is to be represented** in the database & how to access it.
- ▶ Defining Security and Integrity Constraints
 - ↳ DBA **decides on various security and integrity constraints**.
- ▶ Granting of Authorization for Data Access
 - ↳ DBA **determines which user needs access to which part** of the database.
- ▶ Liaison with Users
 - ↳ DBA **provide necessary data** to the user.

Role of DBA

▶ Assisting Application Programmer

↳ DBA **provides assistance to application programmers** to develop application programs.

▶ Monitoring Performance

↳ DBA **ensures that better performance is maintained** by making a change in the physical or logical schema if required.

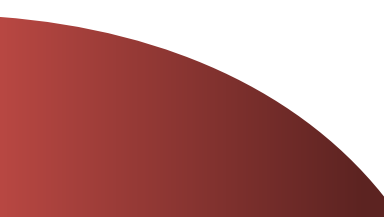
▶ Backup and Recovery

↳ DBA **backing up the database** on some storage devices such as DVD, CD or magnetic tape or remote servers and **recover the system in case of failures**, such as flood or virus attack from this backup.

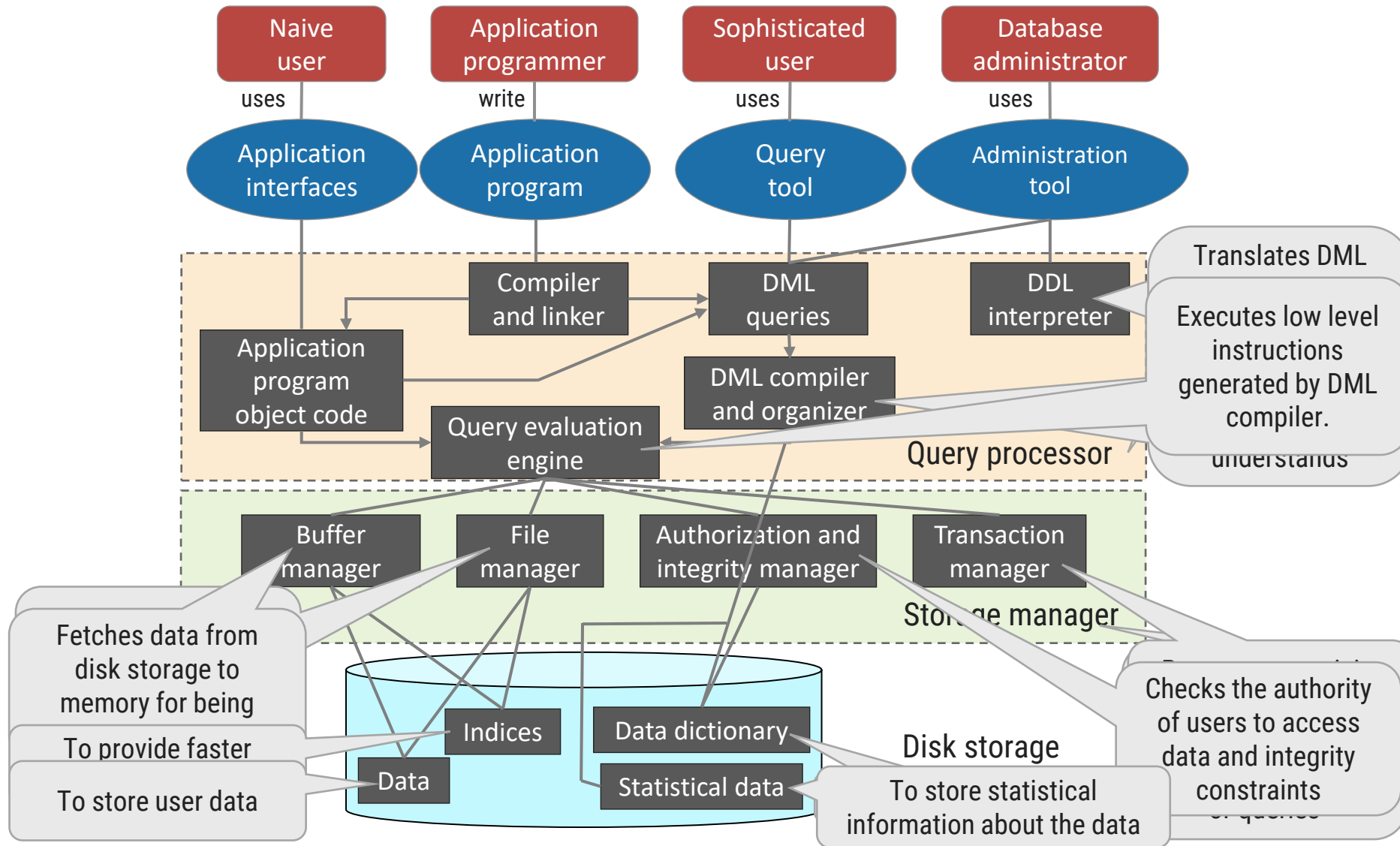


Database System Architecture

Section - 8



Database System Architecture



Questions asked in Exam

1. List and explain the advantages of DBMS over file based system. **OR** Explain disadvantages of files based system.
2. Draw and explain 3 level architecture of DBMS.
3. List and explain different categories/types of database users.
4. List and explain different tasks/roles/functions/duties of DBA (Database Administrator).
5. Explain DBMS architecture with block diagram. **OR** Explain Database System architecture with block diagram.